MULTIMEDIA UNIVERSITY

STUDENT ID NO

MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2024 (TERM ID 2430)

CCP4213 – DATA STRUCTURE AND ALGORITHMS

(All sections / Groups)

20 FEBRUARY 2025

9.00 a.m - 11.00 a.m
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This Question Paper consists of eight (8) pages including cover page.
2. There are five (5) questions. Answer ALL questions. Total is 50 marks.
3. Write your answers in the Answer Booklet provided.

QUESTION 1 [10 Marks]

a) Given the following code segment in Figure 1, answer all of the following questions:

```
int array[ ]={10,20,30,40}, num=50;
int *ptr1, *ptr2;
ptr1 = array;
*(ptr1+1) = *ptr1+1;
ptr2=&num;
*ptr2 = *ptr1+3;
ptr1[3] = array[2];
int &value=num;
*ptr2=value+10;
for(int i=0;i<4;i++)
    cout<<*(ptr1+i)+value<<" ";
```

Figure 1

i. Trace the output. (2 marks)

ii. Briefly explain what happen when the following two statements a) and b) are executed. (2 marks)

a)	`ptr1 = array;`
b)	`int &value = num;`

b) Evaluate the following *postfix* and *prefix* notation. Write the *infix* notation and calculate the result. (2 marks)

i. Postfix : 8 9 3 * 5 – 4 / + 6 2 - +

ii. Prefix : + * / 8 + 2 6 4 – 8 2

Continued .....

c) Referring to Figure 2, *Q* is a *queue* that can store five integers, and *S* is a *stack* that can store four integers. *X* and *Y* are variables that store values 3 and 5 respectively.

Trace and draw the content of *S*, *Q*, *X* and *Y* after each of the following operations/codes are executed.

Drawing must clearly show and use the updated content of *S*, *Q*, *X* and *Y* in each of the subsequent code execution. (2 marks)

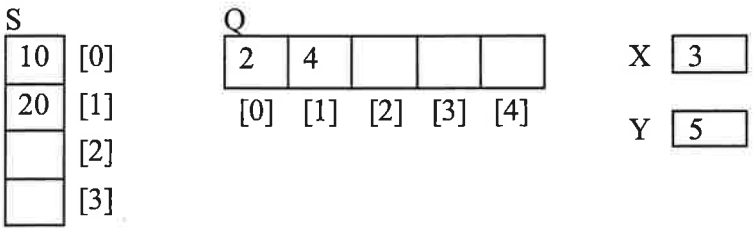

Figure 2

i.	`Q.append(++X + S.pop());`
ii.	`if(Q.serve()+ 2 < Y)` `  S.push((X+Y)%2);`

iii.	`if(S.pop() < Q.serve())`
	`  Q.append(Q.serve()- 3);`
	`else`
	`  S.push(Y++);`

iv.	`Y = S.pop() + Q.serve();`

Continued .....

d) Figure 3 shows an array implementation of a stack. The stack name is *stack*. Variable *top* is used to always store the top element index.

Complete the *pop()* function definition in Figure 4 (part labelled as (I) and (II)) to pop a character value from the *stack*. (2 marks)

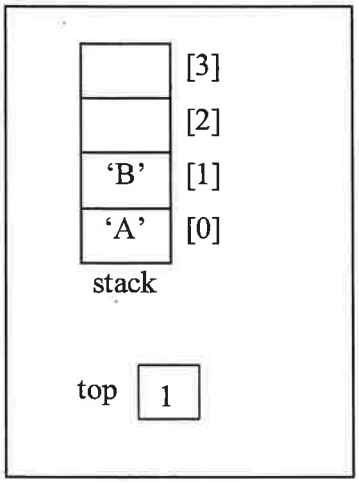

Figure 3

```
char pop()
{
  char grade;
  if (!empty())
  {
   ___________(I)_______________
   ___________(II)______________
   return grade;
  }
  else
  {
   cout<<"Stack is empty."<<endl;
  }
}
```

Figure 4

Continued .....

QUESTION 2 [10 Marks]

Given the following structure and variable declarations in Figure 5 and a linked list diagram in Figure 6, answer all of the following questions.
You can only use variables and pointers declared in Figure 5.
Pointer *head* must always point to the first node in the linked list.

```
struct node{
   int data;
   node *next;
};

node *head, *temp, *prev;
int sum = 0, value = 0;
```

Figure 5

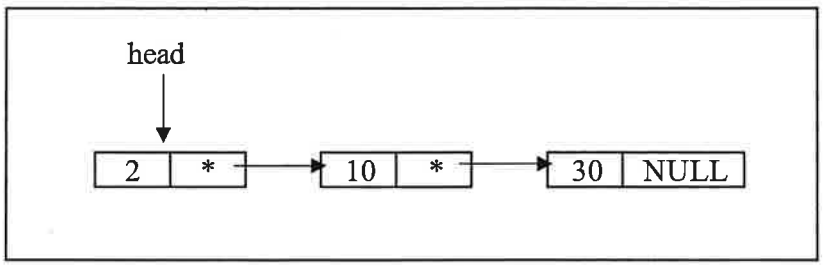

Figure 6

i. Draw an updated diagram after the following lines of code are executed. (2 marks)

```
1. prev = head->next;
2. temp = new node;
3. temp->data = 45;
4. temp->next = prev->next;
5. prev->next = temp;
```

ii. Write code segment to total up all even numbers in the linked list. Pointer *temp* and pointer *head* are currently pointing to the first node. (3 marks)

iii. Write code segment to swap the data value stored in the last node with the data value stored in the first node. Pointer *temp* and pointer *head* are currently pointing to the first node. (5 marks)

Continued .....

QUESTION 3 [10 Marks]

a) Using *binary search* algorithm, show step by step on how to find data with value 40 in the given array in Figure 7. Write the value of *first, last* and *middle* in each step. (3 marks)

10	20	30	40	50	60
[0]	[1]	[2]	[3]	[4]	[5]

Figure 7

b) What is *collision* in hashing? (1 mark)

c) Given the *main()* function in Figure 8, write the function definition for the *hash()* function that truncates the first digits and the last two digits of a six digits number passed as argument from *main()*. The function will display the result after the truncation. You may declare variables as needed in the *hash()* function. (2 marks)

```
int main()
{
    hash(654321);
    return 0;
}
```

Figure 8

d) Assume a hash table with 7 locations and the hashing function $h(x) = x \% 7$. Show the index for each key and the result of the hash table when the following integers are inserted in the order given. Use *Linear Probing* to resolve the collision. (4 marks)

28, 14, 36, 46

Continued .....

QUESTION 4 [10 Marks]

> This sorting algorithm uses the concept of divide and conquer. The list is divided into halves/two repeatedly until all the elements becomes individual. Then, compare the two individual elements, place them in the correct order, ascending or descending, and combine. The process is repeated until all the elements are recompiled as a whole.

a) Based on the description above, what is the name of the sorting algorithm? (1 mark)

b) Figure 9 shows an array of five elements. Show the steps to sort the elements in descending order using *Selection Sort*. (2 marks)

30	20	10	50	40
[0]	[1]	[2]	[3]	[4]

Figure 9

c) Based on the binary tree in Figure 10, answer all the following questions:

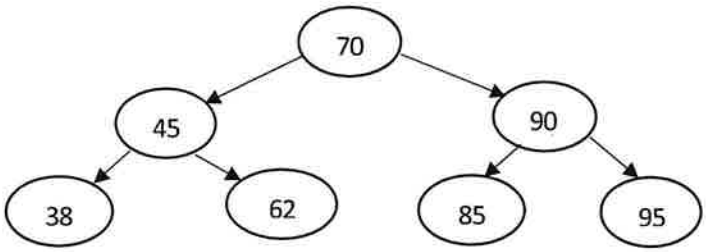

Figure 10

i. What is the height of the tree? (1 mark)

ii. What is the depth of node with data 90? (1 mark)

iii. Write the PreOrder traversal. (1 mark)

iv. Redraw the tree when node with data 50 is inserted and the root node is deleted. (2 marks)

d) Draw a binary tree based on the following algebraic expression: (2 marks)

J / (M + N) – K * L

Continued .....

QUESTION 5 [10 Marks]

a) Given the graph in Figure 11, answer all of the following questions :

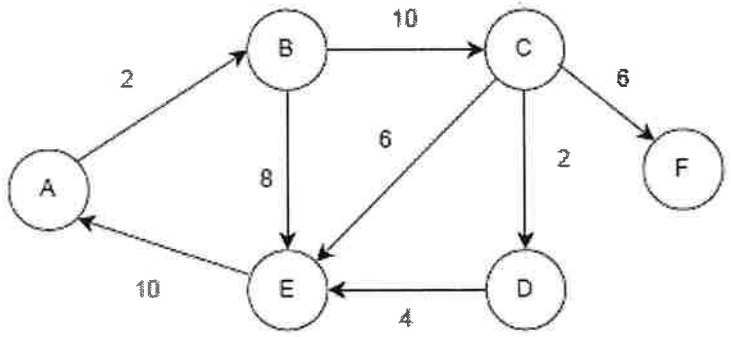

Figure 11

i. Represent the graph using *adjacency matrix*. (3 marks)

ii. State how many cycles can you identify from the graph. Write all the cycle's path and its length. (4 marks)

iii. What is the graph's type? (1 mark)

b) Referring to Figure 12, traverse the graph using *breadth first search* and *depth first search* starting at vertex A. (2 marks)

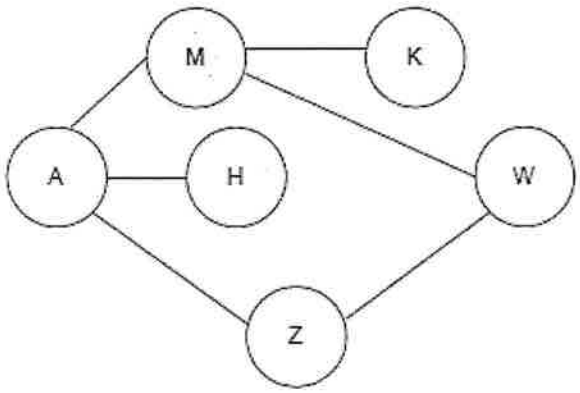

Figure 12

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